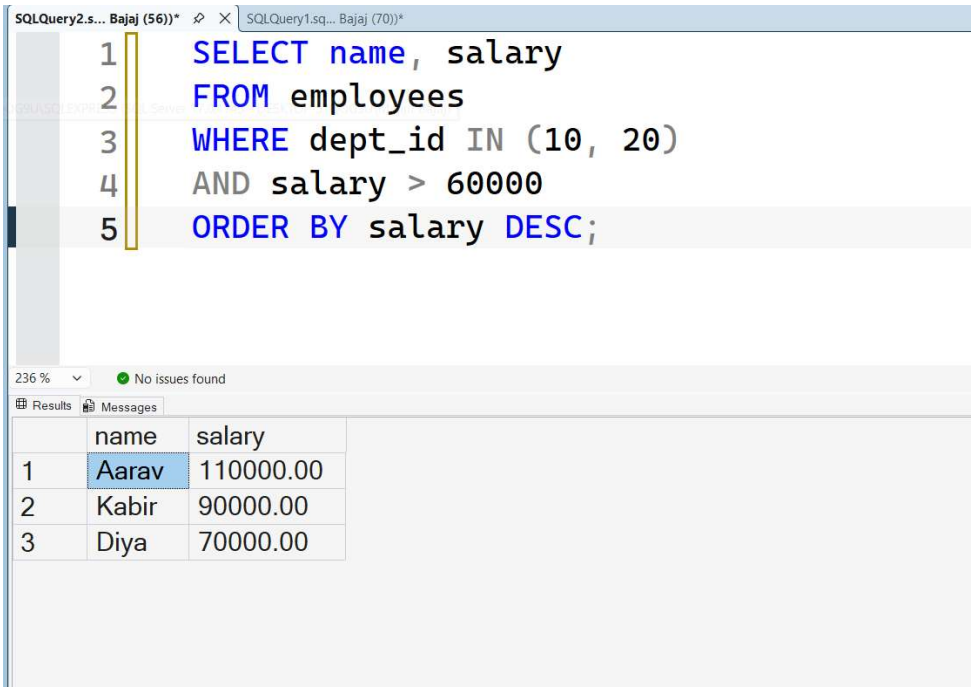
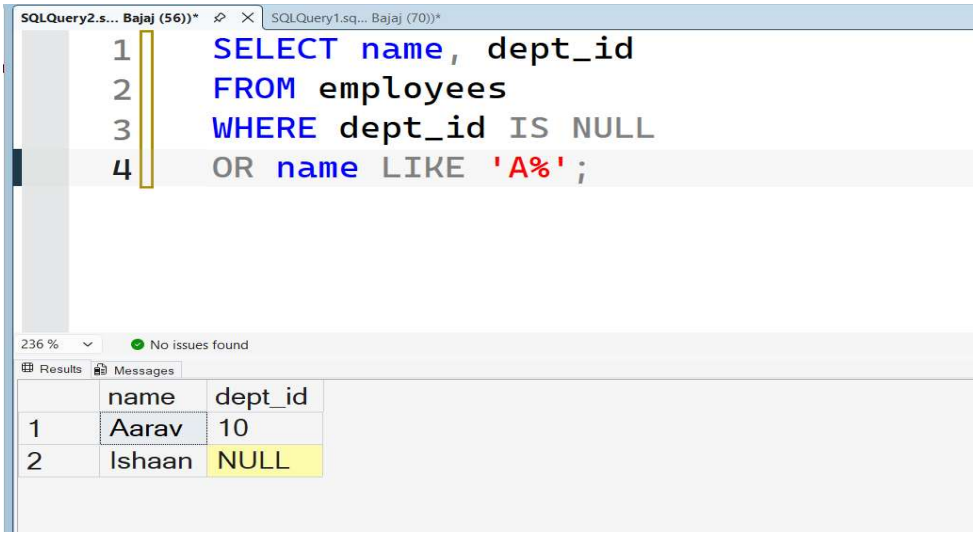
Celebal Learning Engagement Assignments
Week 2 - SQL Queries

SQL Practical Exercises

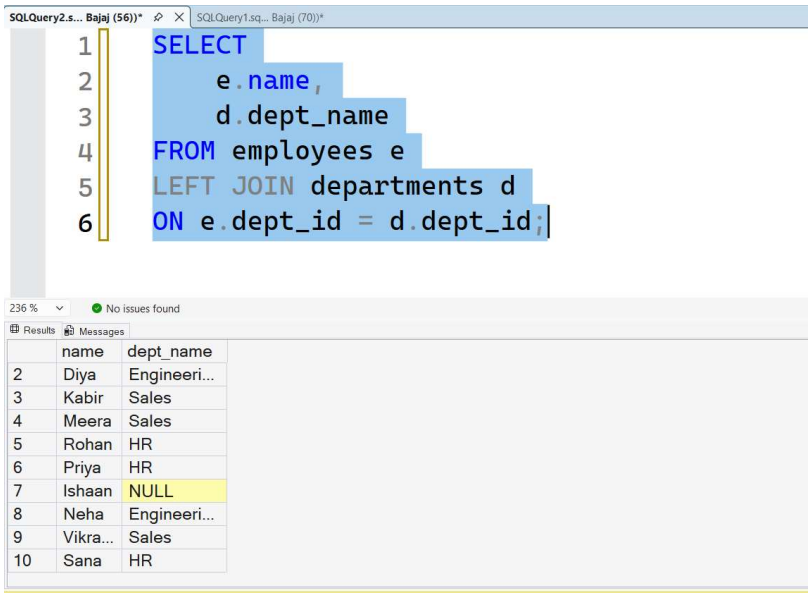
Q1. Retrieve employees from departments 10 and 20 earning more than 60,000, sorted by salary in descending order.



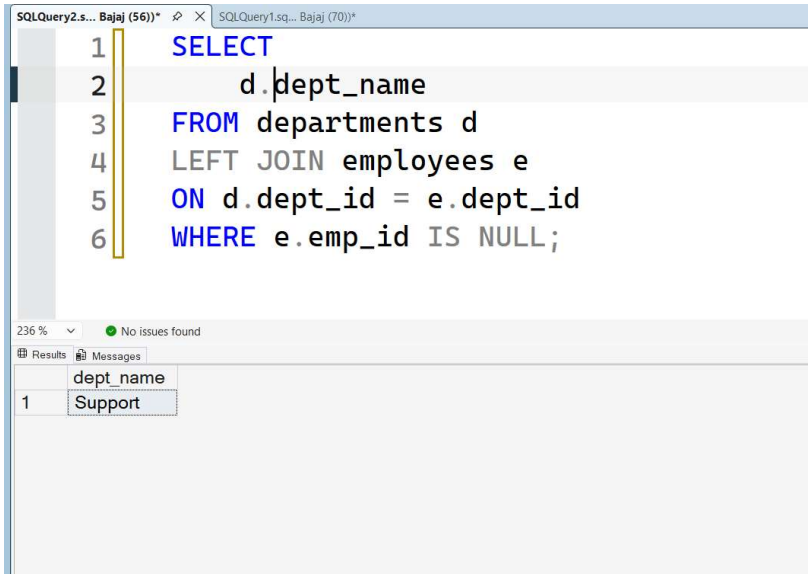
Q2. Find employees who do not belong to any department or whose names start with 'A'.



Q3. Display all employees along with their department names, including employees who do not belong to any department.



Q4. Find departments that do not have any employees.



Q5. Display department names, employee headcount, and average salary for each department.

SQLQuery2.s... Bajaj (56))* | SQLQuery1.sq... Bajaj (70))*

```sql
SELECT
    d.dept_name,
    COUNT(*) AS headcount,
    AVG(e.salary) AS avg_salary
FROM employees e
INNER JOIN departments d
ON e.dept_id = d.dept_id
GROUP BY d.dept_name;
```

195 % No issues found

Results Messages

	dept_name	headcount	avg_salary
1	Engineering	3	80000.000000
2	HR	3	55000.000000
3	Sales	3	65000.000000

Q6. Find departments whose average salary is greater than 60,000.

SQLQuery2.s... Bajaj (56))* | SQLQuery1.sq... Bajaj (70))*

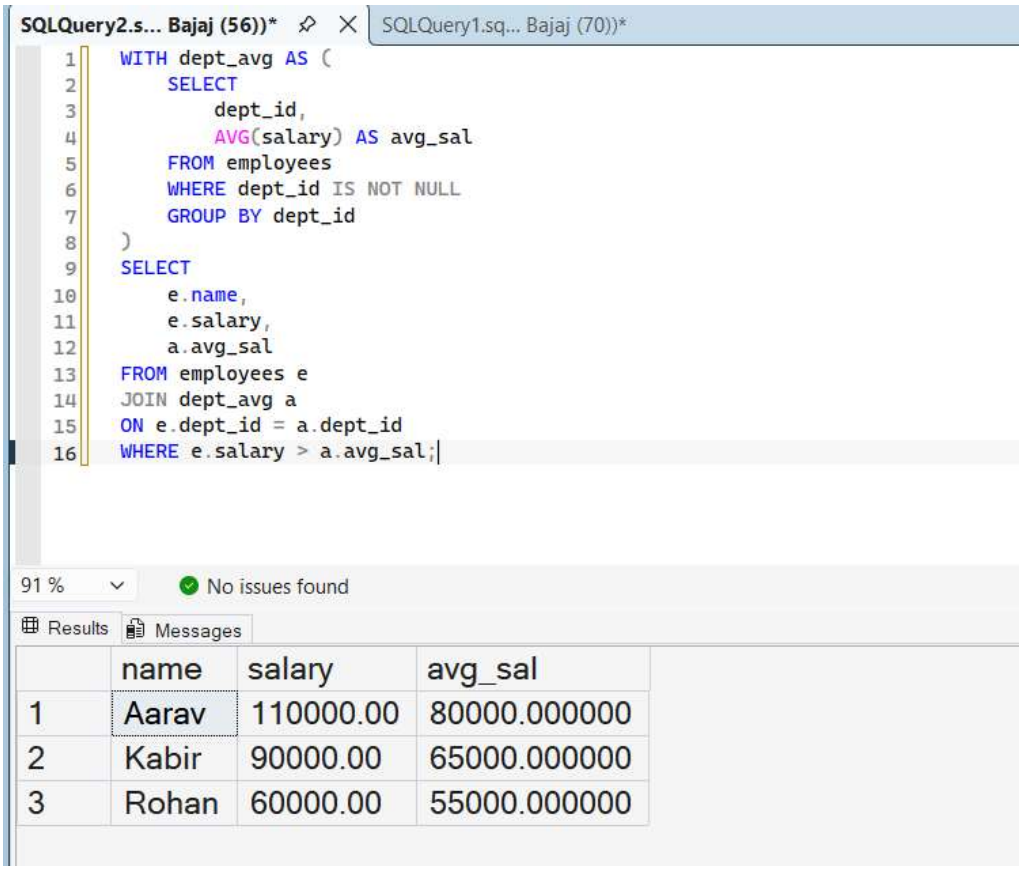
```sql
SELECT
    dept_id,
    AVG(salary) AS avg_salary
FROM employees
WHERE dept_id IS NOT NULL
GROUP BY dept_id
HAVING AVG(salary) > 60000;
```

195 % No issues found

Results Messages

	dept_id	avg_salary
1	10	80000.000000
2	20	65000.000000

Q7. Find employees whose salary is greater than their department's average salary.



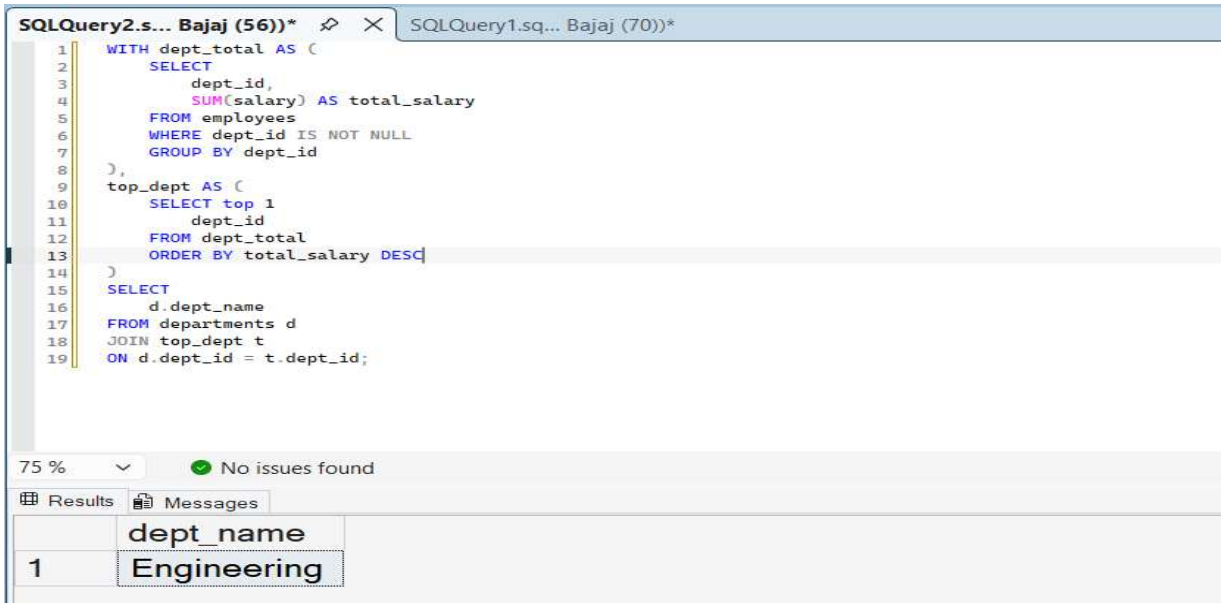
```
SQLQuery2.s... Bajaj (56))*   SQLQuery1.sq... Bajaj (70))*
1  WITH dept_avg AS (
2      SELECT
3          dept_id,
4          AVG(salary) AS avg_sal
5      FROM employees
6      WHERE dept_id IS NOT NULL
7      GROUP BY dept_id
8  )
9  SELECT
10     e.name,
11     e.salary,
12     a.avg_sal
13 FROM employees e
14 JOIN dept_avg a
15 ON e.dept_id = a.dept_id
16 WHERE e.salary > a.avg_sal;
```

91 % No issues found

Results Messages

	name	salary	avg_sal
1	Aarav	110000.00	80000.000000
2	Kabir	90000.00	65000.000000
3	Rohan	60000.00	55000.000000

Q8. Find the department that spends the most on employee salaries.



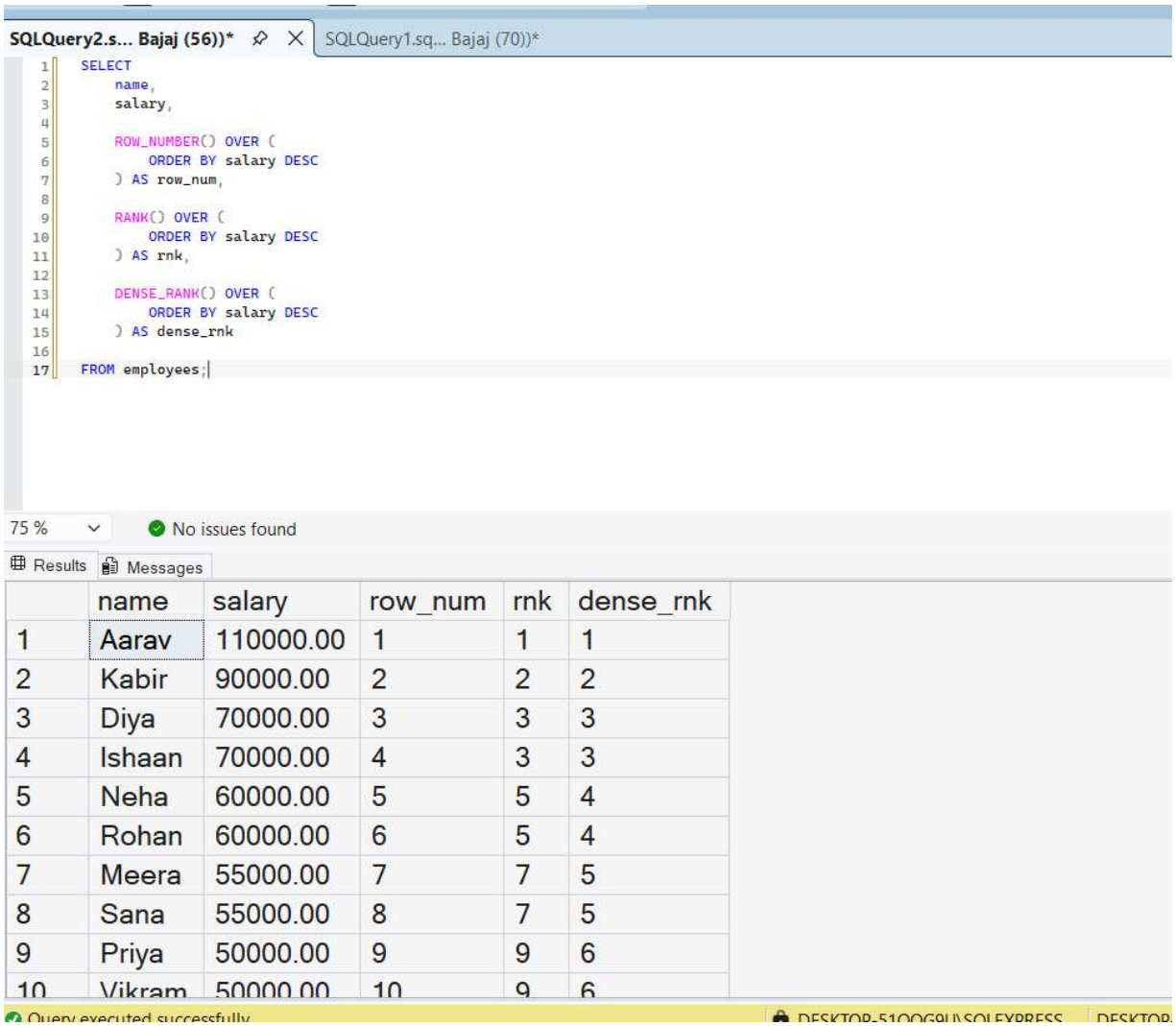
```
SQLQuery2.s... Bajaj (56))*   SQLQuery1.sq... Bajaj (70))*
1  WITH dept_total AS (
2      SELECT
3          dept_id,
4          SUM(salary) AS total_salary
5      FROM employees
6      WHERE dept_id IS NOT NULL
7      GROUP BY dept_id
8  ),
9  top_dept AS (
10     SELECT top 1
11         dept_id
12     FROM dept_total
13     ORDER BY total_salary DESC
14 )
15 SELECT
16     d.dept_name
17 FROM departments d
18 JOIN top_dept t
19 ON d.dept_id = t.dept_id;
```

75 % No issues found

Results Messages

	dept_name
1	Engineering

Q9. Display employee salaries along with their row number, rank, and dense rank.



SQLQuery2.s... Bajaj (56))* | SQLQuery1.sq... Bajaj (70))*

```
SELECT
    name,
    salary,

    ROW_NUMBER() OVER (
        ORDER BY salary DESC
    ) AS row_num,

    RANK() OVER (
        ORDER BY salary DESC
    ) AS rnk,

    DENSE_RANK() OVER (
        ORDER BY salary DESC
    ) AS dense_rnk

FROM employees;
```

75 % | No issues found

Results | Messages

	name	salary	row_num	rnk	dense_rnk
1	Aarav	110000.00	1	1	1
2	Kabir	90000.00	2	2	2
3	Diya	70000.00	3	3	3
4	Ishaan	70000.00	4	3	3
5	Neha	60000.00	5	5	4
6	Rohan	60000.00	6	5	4
7	Meera	55000.00	7	7	5
8	Sana	55000.00	8	7	5
9	Priya	50000.00	9	9	6
10	Vikram	50000.00	10	9	6

Query executed successfully | DESKTOP-51OOG9U\SQLEXPRESS | DESKTOP...

Q10. Find the highest-paid employee in each department.

SQLQuery2.s... Bajaj (56))* | SQLQuery1.sq... Bajaj (70))*

```sql
WITH ranked AS (
    SELECT
        name,
        dept_id,
        salary,

        DENSE_RANK() OVER (
            PARTITION BY dept_id
            ORDER BY salary DESC
        ) AS r

    FROM employees
)
SELECT
    name,
    dept_id,
    salary
FROM ranked
WHERE r = 1;
```

75 % | No issues found

Results | Messages

	name	dept_id	salary
1	Ishaan	NULL	70000.00
2	Aarav	10	110000.00
3	Kabir	20	90000.00
4	Rohan	30	60000.00

Conclusion

Through these practical exercises, I developed an understanding of SQL fundamentals and advanced querying techniques. The exercises provided hands-on experience in working with relational databases, retrieving meaningful insights, performing data aggregations, and analysing records using joins, CTEs, and window functions.

These concepts form an important foundation for data analysis, data engineering, and working with large-scale datasets.